

Electrical and Electronics Engineering and Robotics

ELEC345 High Speed Communications

Smith Charts and L Matching networks

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$Z_0 = 50\Omega$
 200Ω

On chart $\frac{200}{50} = 4 = Z_L$

$Z_0 = 50\Omega$
 $Z_L = 25 + 25j$
 $\div 50 \Rightarrow 0.5 + 0.5j$

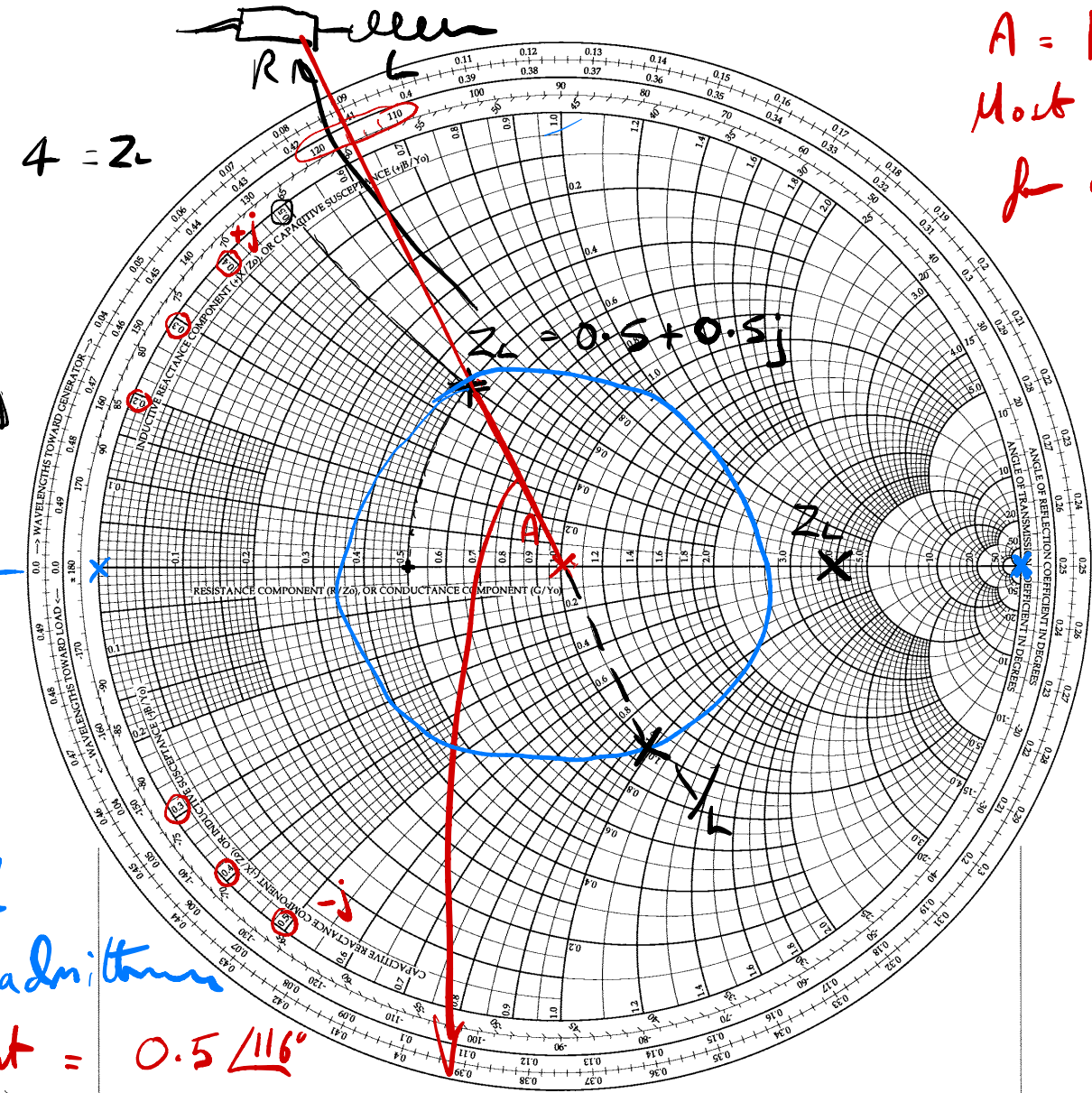
$A = 1 = Z_0$
 Most of the time
 for us = 50Ω

Short
Circuit

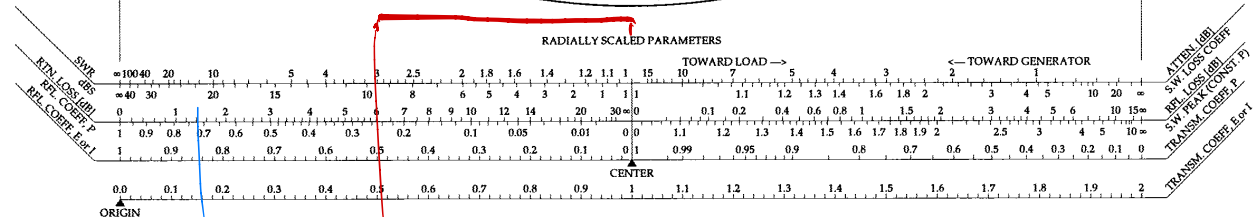
Open
Circuit

x axis is real
 impedance or admittance

Reflection coefficient = $0.5 \angle 116^\circ$

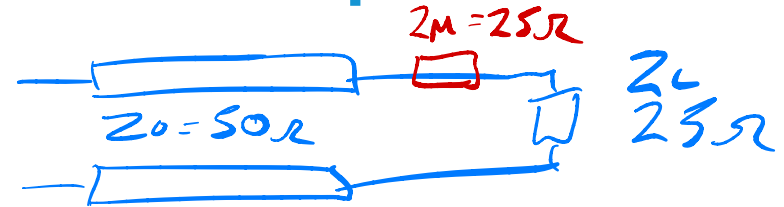


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Reactive, frequency selective components

- A bit of basic theory/background:
 - Impedance, Z (ohms, Ω)



	Symbol	Component Value	Impedance at 1kHz	Impedance at 1MHz		Admittance at 1MHz (1/Z)
$Z_R = R$		10 Ω	10 Ω	10 Ω	$Y = 1/R$	0.1S
$Z_L = j\omega L$		10mH	$j62\Omega$	$j62k\Omega$	$Y = \cancel{j} / j\omega L$ $\times j$	$-j16\mu S$
$Z_C = 1/j\omega C$		10nF	$-j16k\Omega$	$-j15\Omega$	$Y = j\omega C$	0.066S

Where $\omega = 2\pi f$ and f = frequency.

	s	p
$\dot{H}$	$-j$	$+j$
m	$+j$	$-j$

$$- \frac{j}{j\omega C}$$


jwL

jwc



= $+j$ up the smith chart

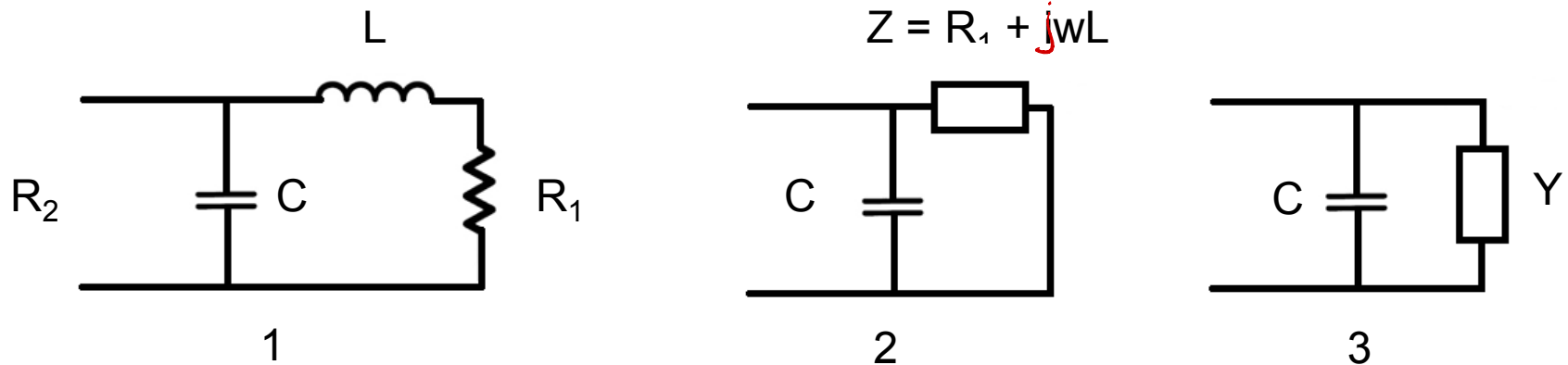


-j down the smith chart

Z to Y & vice versa mirror though the centre of the chart.

L Match Design Equations

1. We wish to provide a L match to transform R_1 to R_2



2. First, look at the series part $\Rightarrow Z = R_1 + j\omega L$
3. Take its parallel form because we have the parallel C.

$$Y = \frac{1}{Z} = \frac{1}{R_1 + j\omega L} = \frac{1}{R_1 + j\omega L} * \frac{R_1 - j\omega L}{R_1 - j\omega L} = \frac{R_1 - j\omega L}{R_1^2 + \omega^2 L^2}$$

L Match Design Equations

The real part must equal the real part of the value we're matching too, in this case $1/R_2$ (parallel) or R_2 (series)

Our inductor and R_1 will have a value of $A-jB$

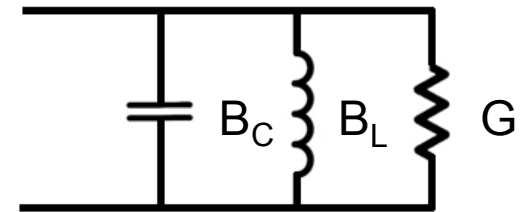
$$Y = A - jB = \frac{R_1}{R_1^2 + \omega^2 L^2} - \frac{j\omega L}{R_1^2 + \omega^2 L^2}$$

$$\frac{1}{R_2} = \frac{R_1}{R_1^2 + \omega^2 L^2}$$

$$R_1^2 + \omega^2 L^2 = R_1 R_2$$

$$\omega^2 L^2 = R_1 R_2 - R_1^2$$

$$\omega^2 L^2 = R_1(R_2 - R_1)$$



$$L = \sqrt{R_1(R_2 - R_1) / \omega^2}$$

$$B_L = -\omega L / R_1 R_2$$

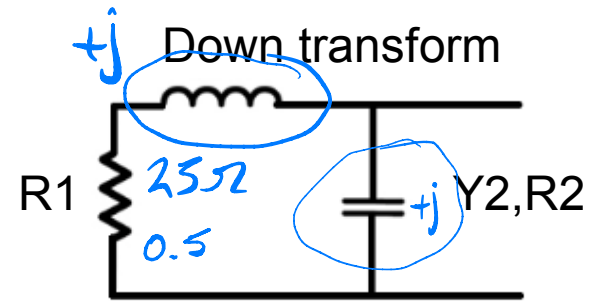
$$B_C = \omega L / R_1 R_2$$

$$C = L / R_1 R_2$$

L Match Design Equation

Resistive example R1, R2

Inductor = +j


$$Z_0 = R_L = 50\Omega$$

- $R_2 = 250\Omega$

Impedance

1. Normalize (/50) $\frac{25}{50} = 0.5$

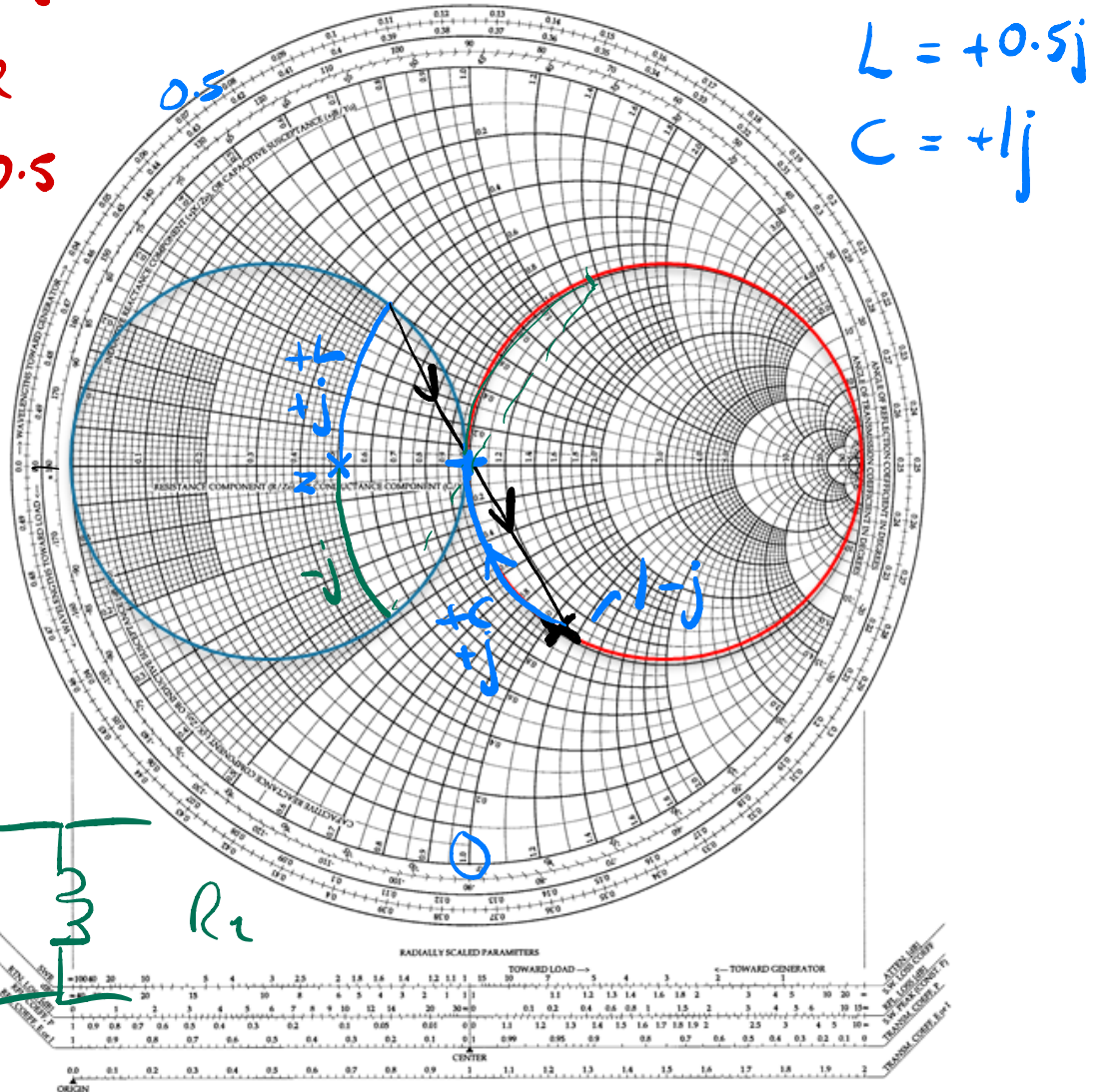
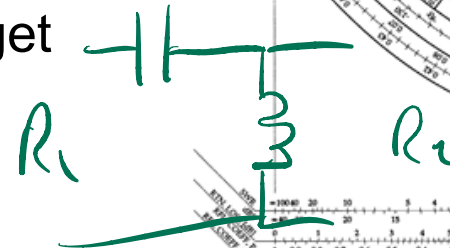
1. $z_1=1$ (Centre)

~~2. $x_2 = 5, y_2 = 0.2$~~

2. Find points on matching circles

3. Start from centre
& join the dots

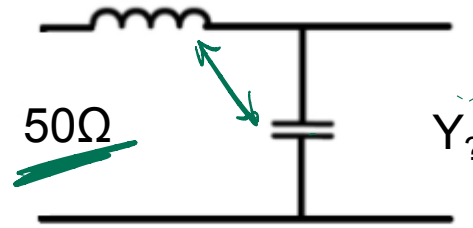
4. Read values off
& de-normalize to get
Component values



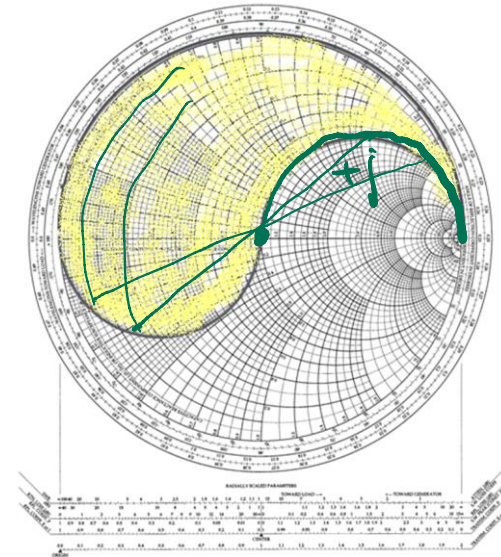
What match does the network provide?

What values can we match?

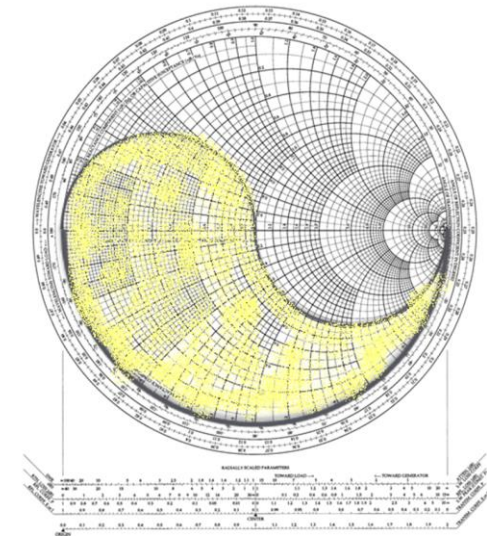
1. First normalize, our starting point is 50 which becomes 1 at the centre of the chart
2. Series inductor, $+j$ will take us anywhere along the top half of the matching circle in impedance, Z .
3. Transpose through the centre to Y (admittance) on the inverse matching circle
4. Add $+j$ for the parallel C
5. Transpose through the centre to the Z match.



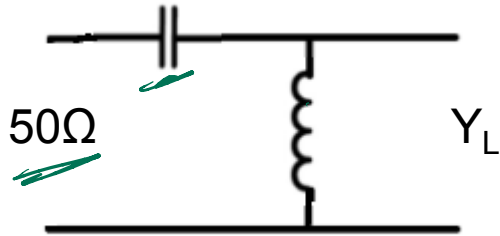
The admittance provided by the network.
The complex conjugate to the load.



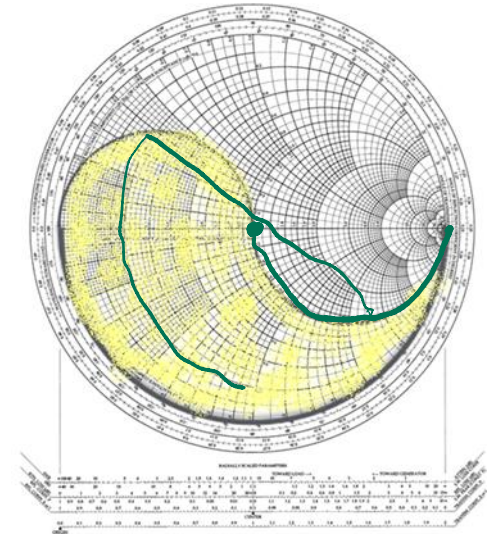
The admittances that can be matched.



L Match, down transform, high pass



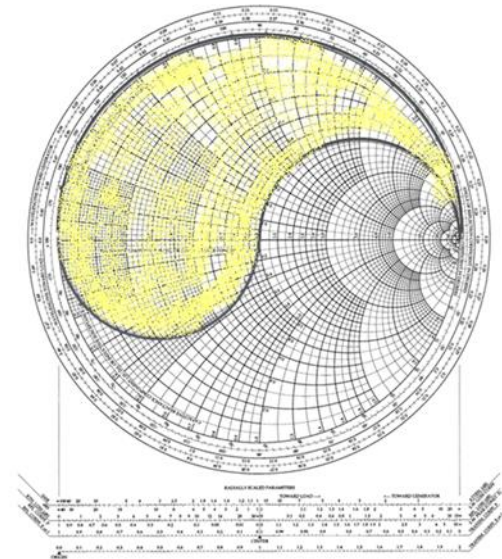
Match provided by
the network



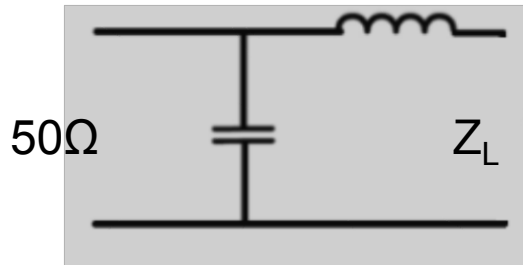
What values can we match?

1. First normalize, our starting point is 50 which becomes 1 at the centre of the chart
2. Series C, $-j$ will take us anywhere along the bottom half of the matching circle, Z
3. Transpose through the centre to Y on the inverse matching circle
4. $-j$ for the parallel L
5. Transpose through the centre to the Z match.

The load it matches
to.



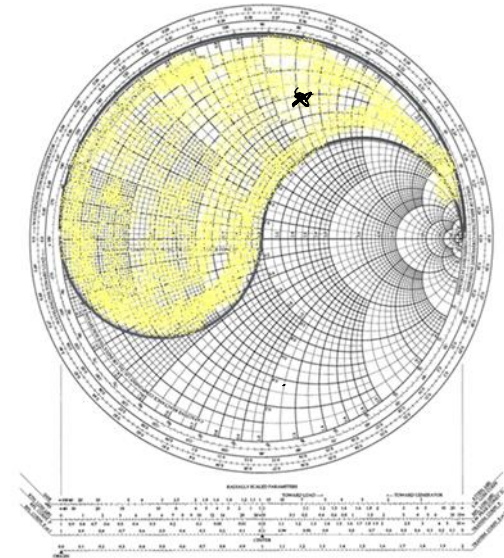
L Match, up transform, low pass



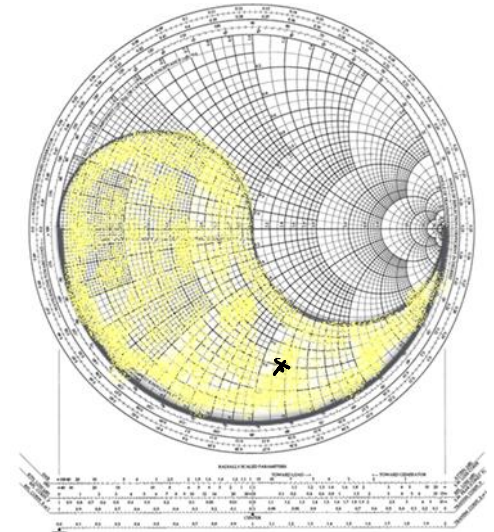
Match provided by the network

What values can we match?

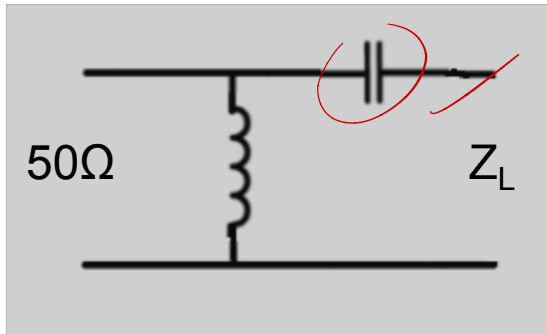
1. First normalize, our starting point is 50 which becomes 1 at the centre of the chart
2. Parallel capacitor, $+j$ will take us anywhere along the top half of the matching circle, Y
3. Transpose through the centre to Z on the inverse matching circle
4. Add $+j$ for the series L to Z match



The load it matches to.



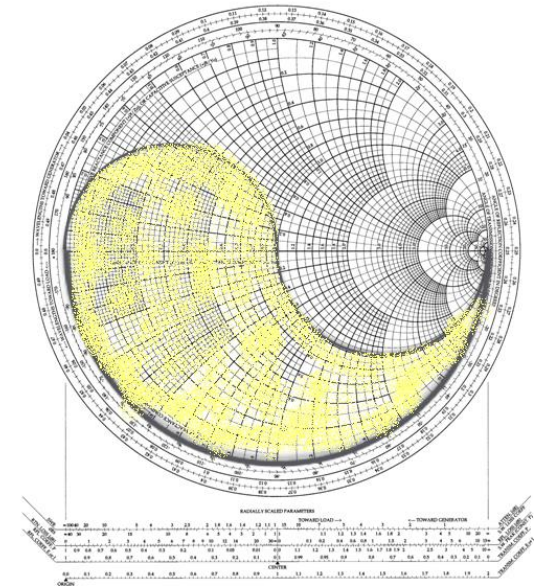
L Match, up transform, high pass



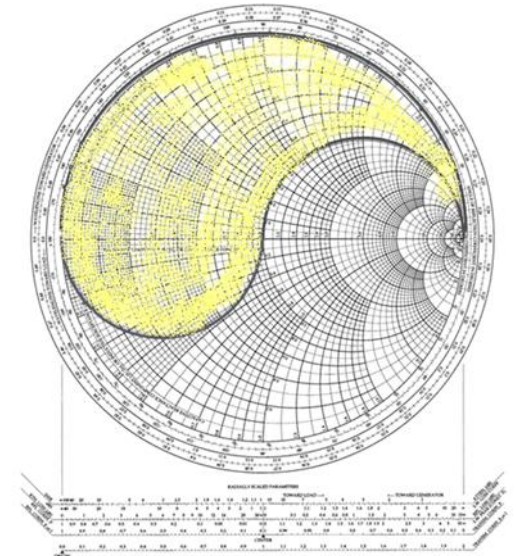
Match provided by
the network

What can we match to?

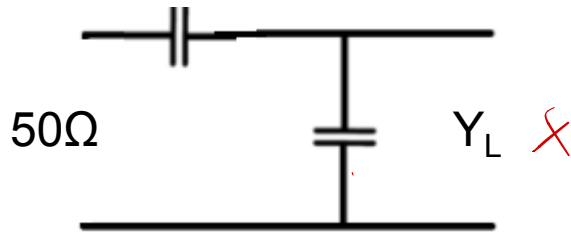
1. First normalize, our starting point is 50 which becomes 1 at the centre of the chart
2. Parallel inductor, $-j$ will take us anywhere along the bottom half of the matching circle, Y
3. Transpose through the centre to Z on the inverse matching circle
4. $-j$ for the series C and Z match.



The load it matches
to.



L Match

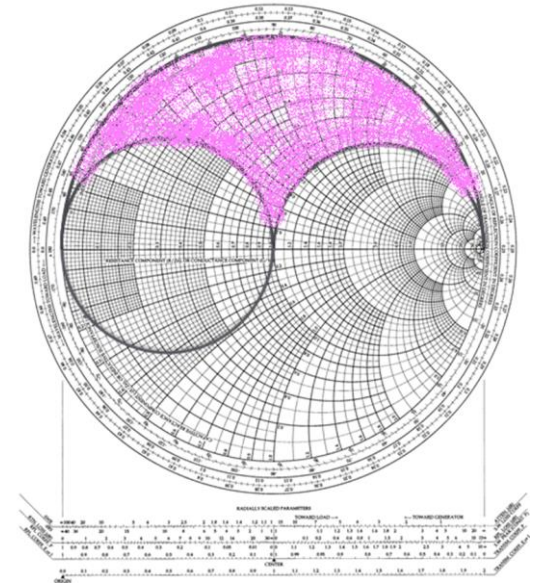


Match provided by
the network

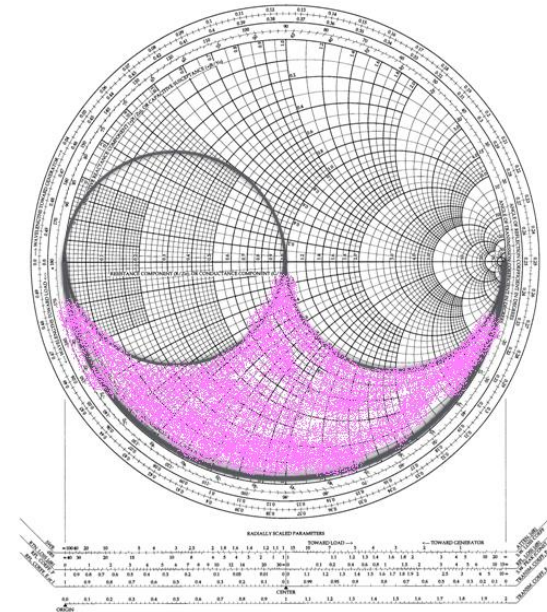
What can we match to?

1. First normalize, our starting point is 50 which becomes 1 at the centre of the chart
2. Series capacitor, $-j$ will take us anywhere along the bottom half of the matching circle, Z
3. Transpose through the centre to Y on the inverse matching circle
4. Add $+j$ for the parallel C
5. Transpose through the centre to the Z match.

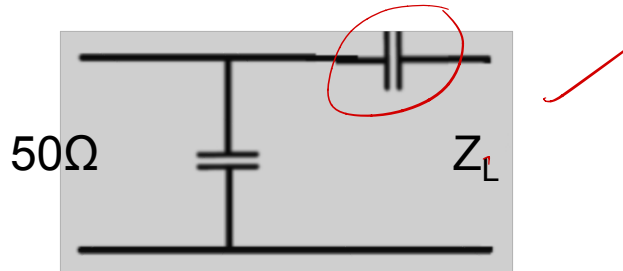
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to.



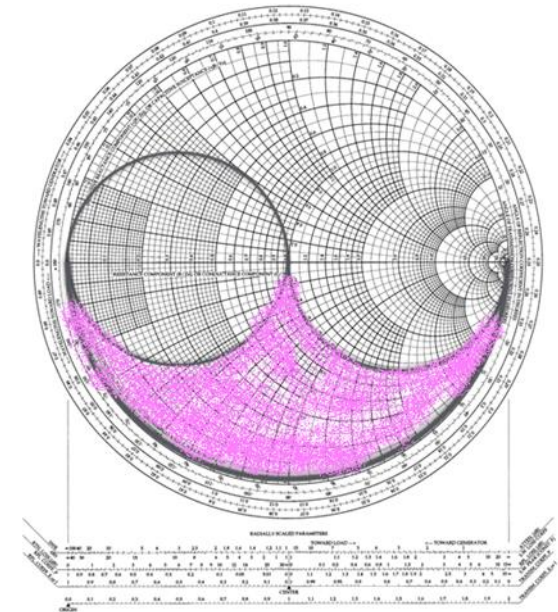
L Match



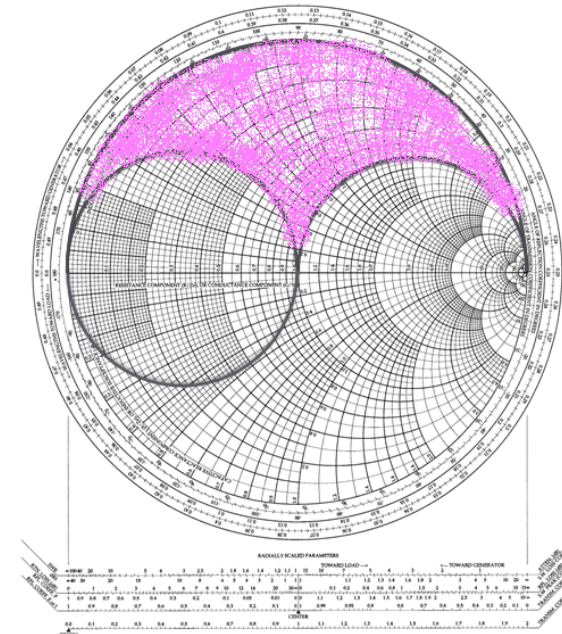
Match provided by
the network

What can we match to?

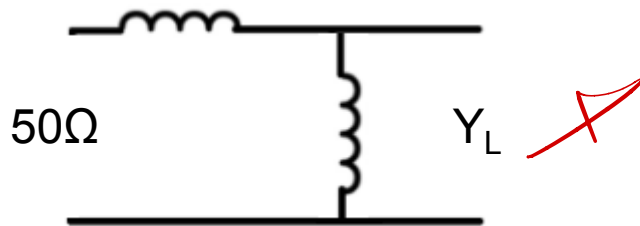
1. First normalize, our starting point is 50 which becomes 1 at the centre of the chart
2. Parallel capacitor, $+j$ will take us anywhere along the top half of the matching circle, Y
3. Transpose through the centre to Z on the inverse matching circle
4. $-j$ for the series C and Z match.



The load it matches
to.



L Match

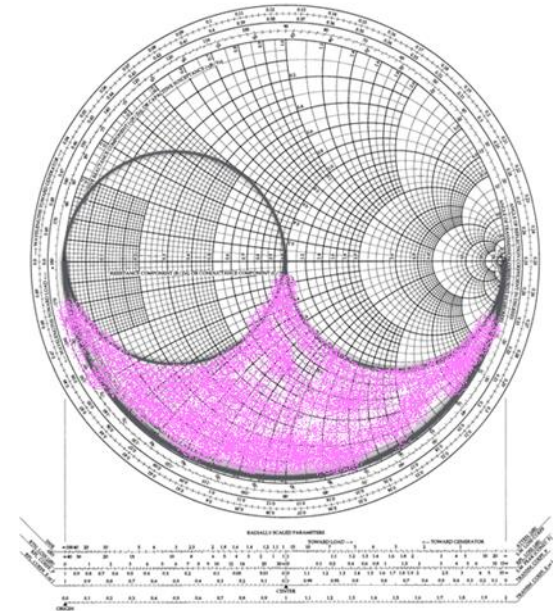


Match provided by
the network

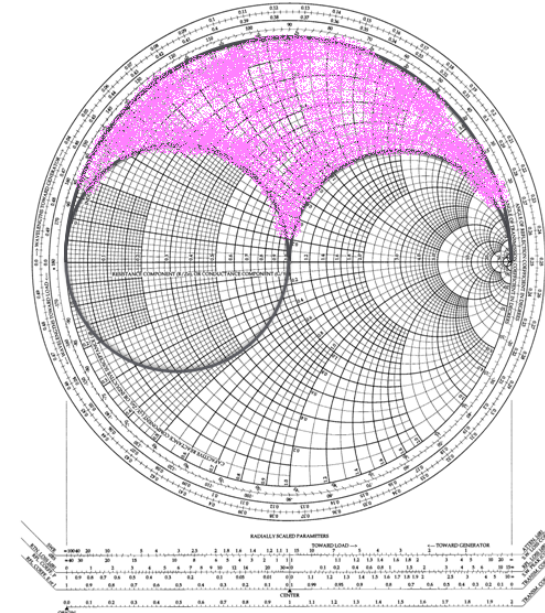
What can we match to?

1. First normalize, our starting point is 50 which becomes 1 at the centre of the chart
2. Series inductor, $+j$ will take us anywhere along the top half of the matching circle, Z
3. Transpose through the centre to Y on the inverse matching circle
4. $-j$ for the parallel L
5. Transpose through the centre to the Z match.

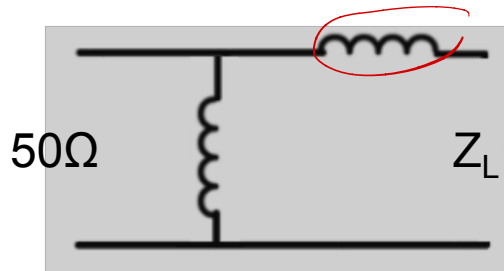
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The load it matches
to.



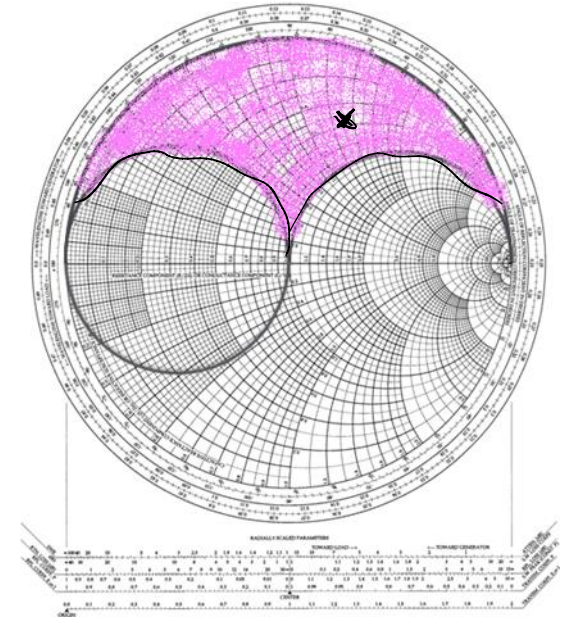
L Match



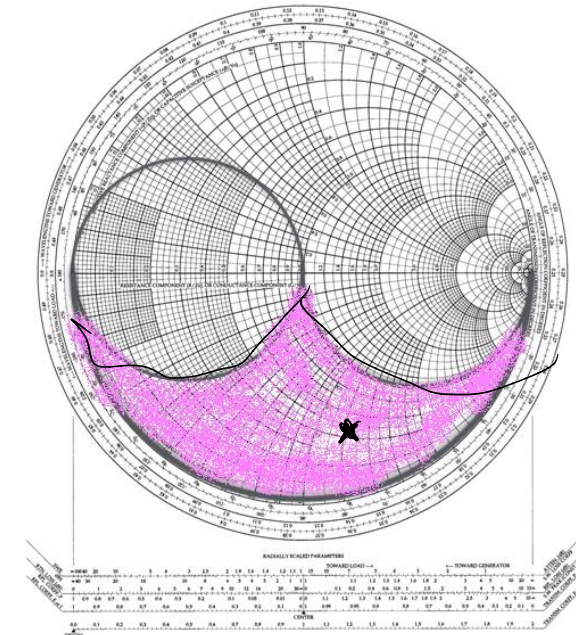
Match provided by
the network

What can we match to?

1. First normalize, our starting point is 50 which becomes 1 at the centre of the chart
2. Parallel inductor, $-j$ will take us anywhere along the top half of the matching circle, Y
3. Transpose through the centre to Z on the inverse matching circle
4. Add $+j$ for the series L and Z match.



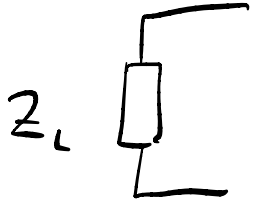
The load it matches
to.



$$Z_0 = 50$$

Match Z_L to Z_0

$$Z_L = 10 + 90j \Omega$$



Z_0
%

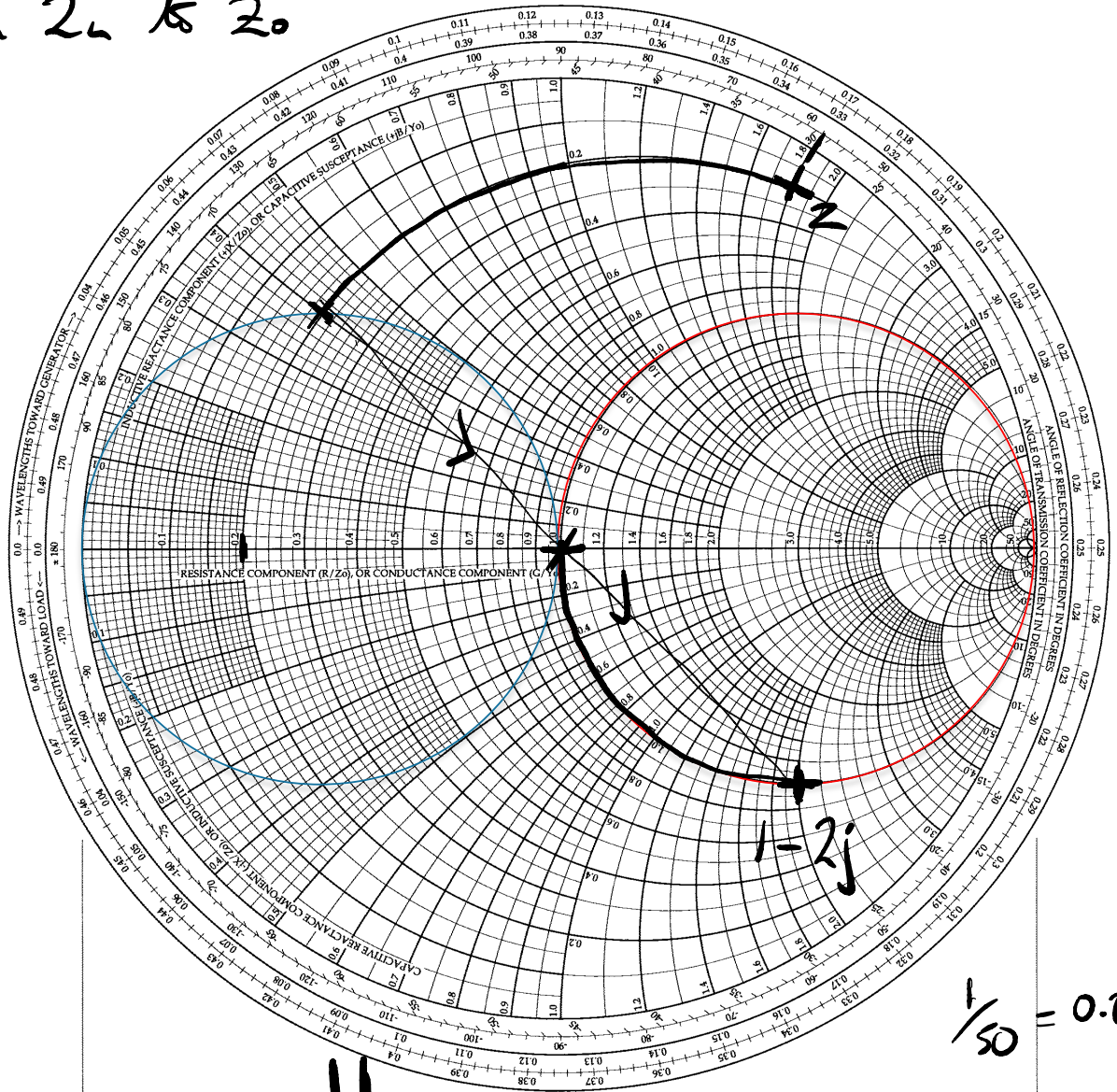
① Normalize ($\div Z_0$) = $0.2 + 1.8j$

② Plot

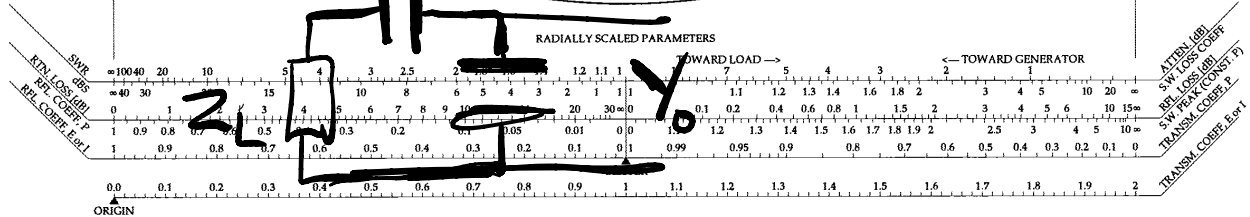
③ $\pm j$ to guide circle $\begin{matrix} @ +1.8j \\ -1.4j \end{matrix}$ $\begin{matrix} 1 \\ +0.4j \end{matrix}$

④ Transpose

⑤ $\pm j$ to remove left over j
@ $1 - 2j$ I need $+2j$
to get to the cube



$$\frac{1}{50} = 0.02$$



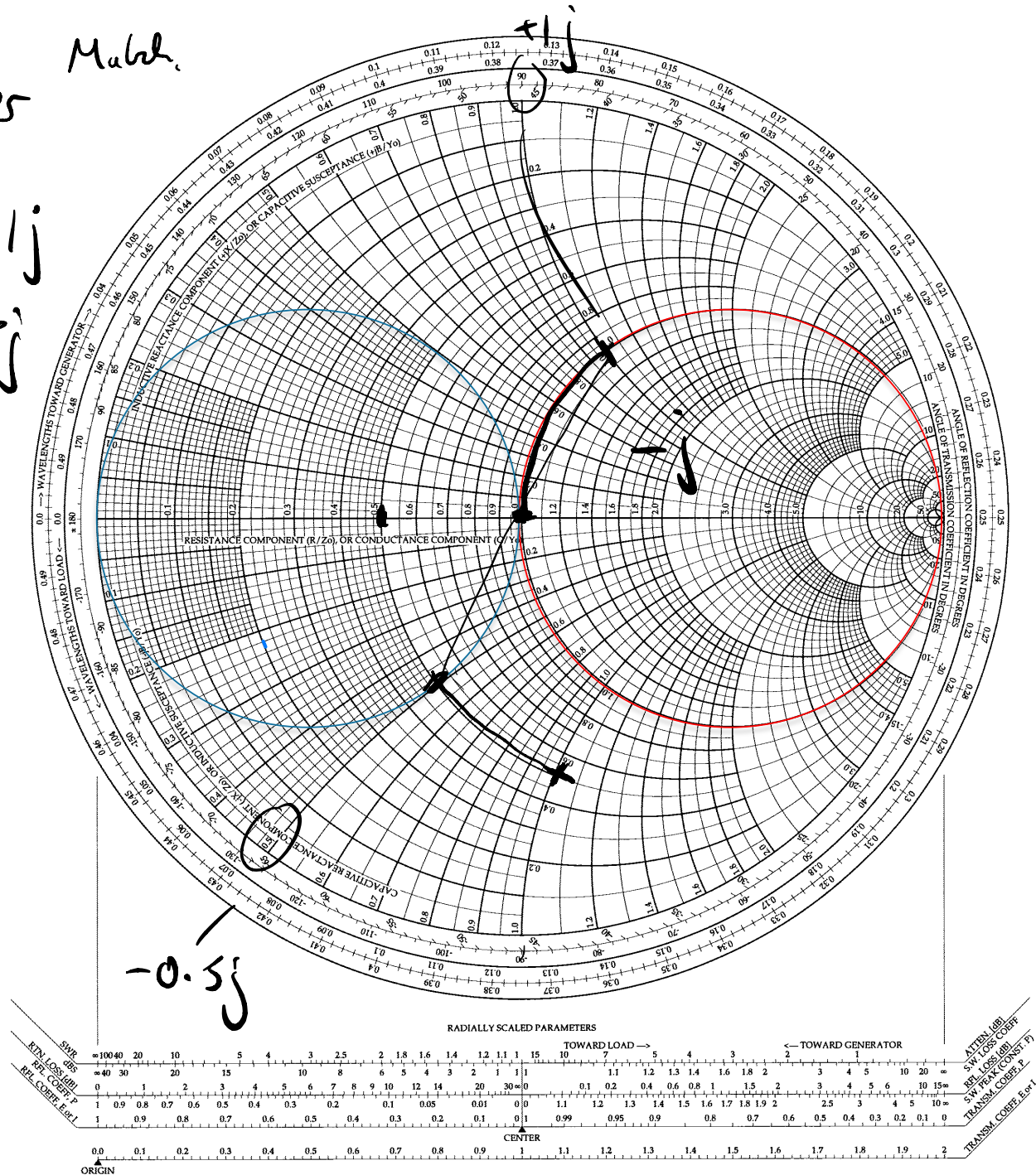
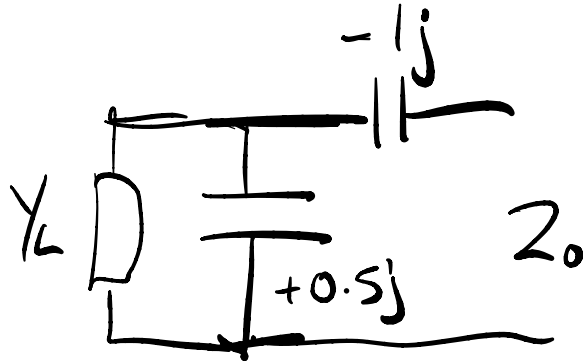
$$Z_0 = 50 \Omega$$

$$Y_L = 0.01 - 0.02j \text{ S}$$

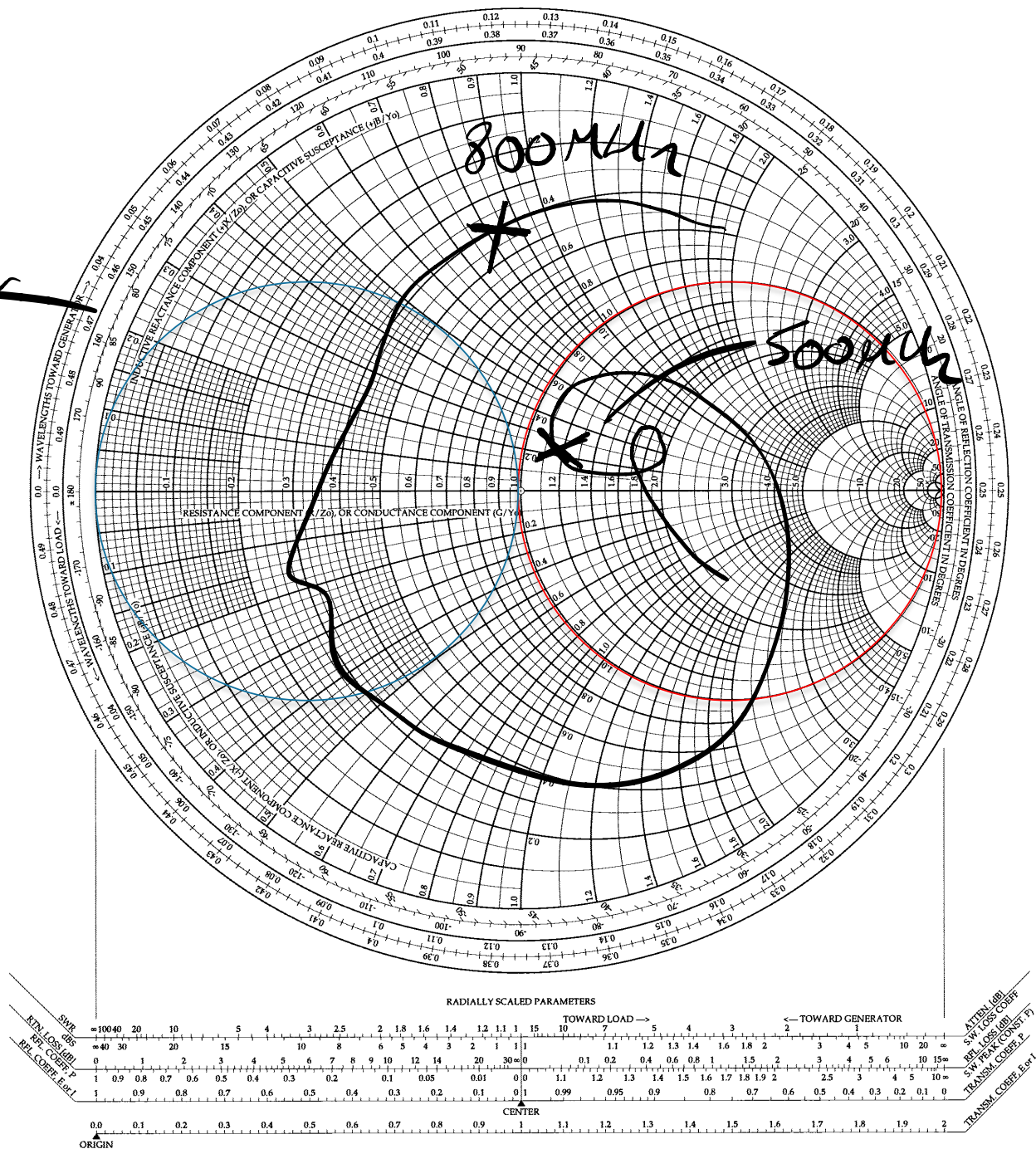
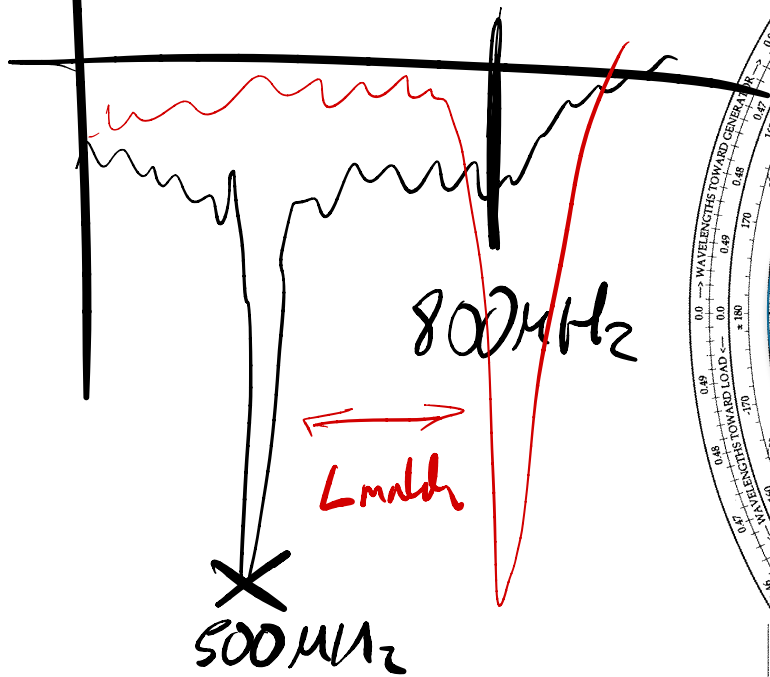
Math.

$$\textcircled{1} \text{ Normalize } (x Z_0) = 0.5 - 1j$$

$$\textcircled{2} \text{ Get to circle } +0.5j$$



VNA



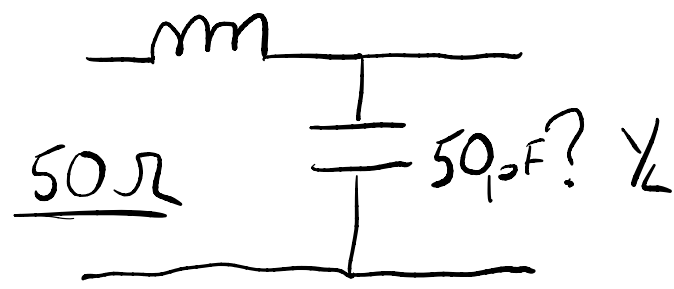
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$Z_0 = 50 \Omega$

@ 200 MHz

60 nH

$\omega = 2 \times \pi \times f$



$$X_L = j\omega L$$

$$= j \cdot 2 \cdot \pi \times 200 \times 10^6 \times 60 \times 10^{-9}$$

$$= 75.4 j \Omega$$

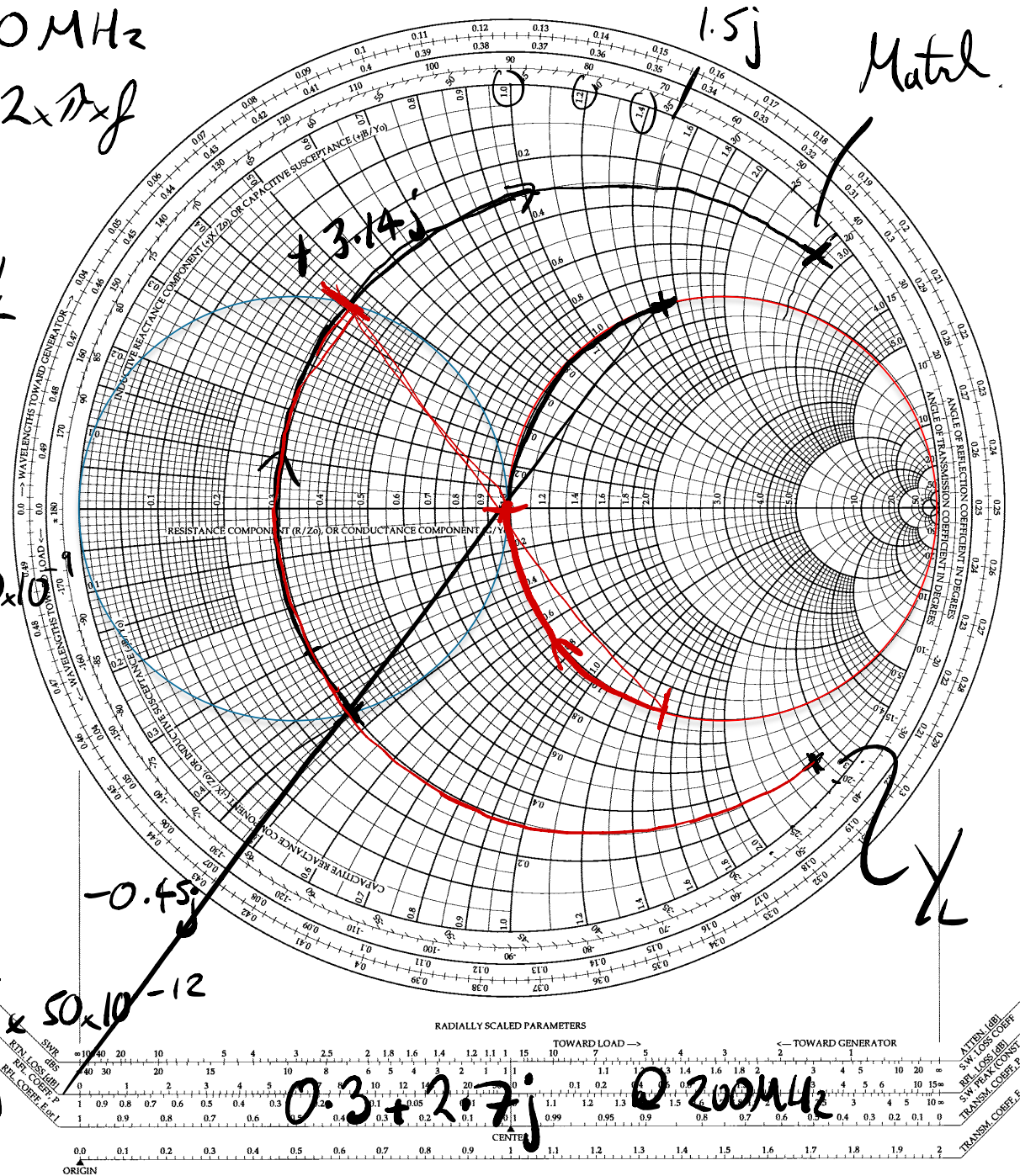
$\frac{Z_0}{Z_L} \Rightarrow \frac{75.4 j}{50} = 1.5 j$

$Y_C = j\omega C$

$$j \cdot 2 \times \pi \times 200 \times 10^6 \times 50 \times 10^{-12}$$

$$= 0.063 j$$

Admittance ($\times 50$) = $3.14 j$

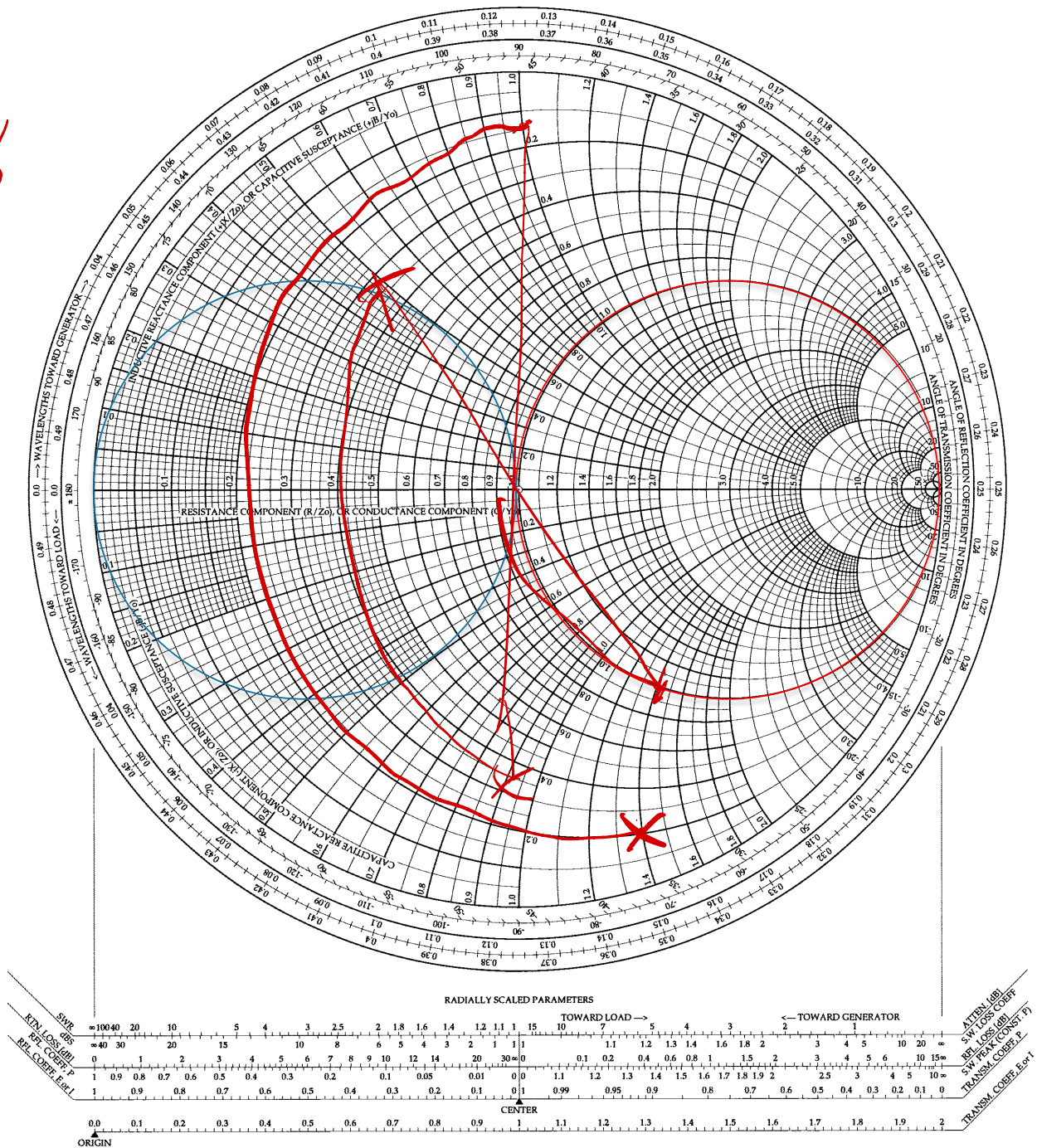
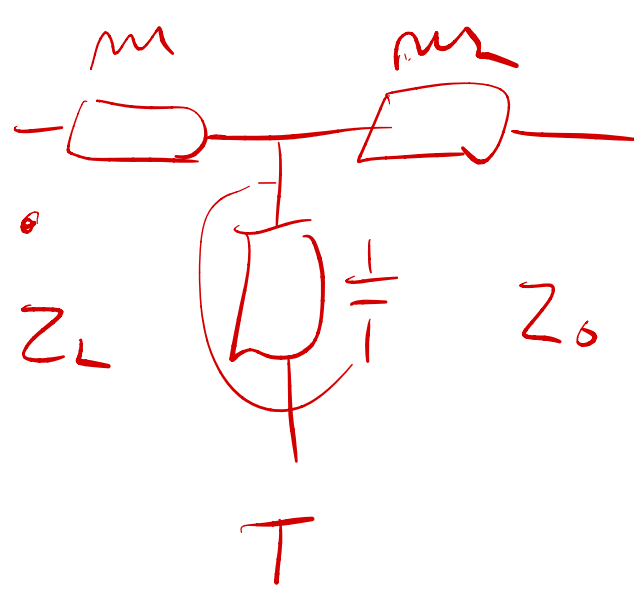
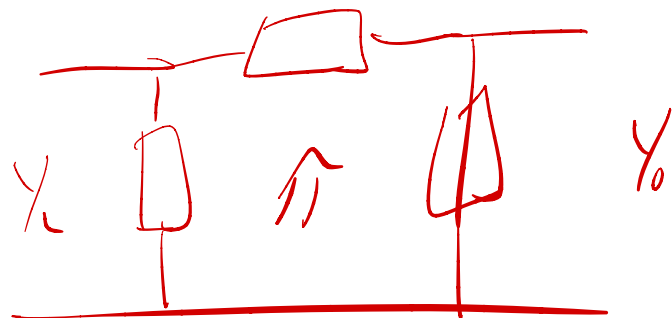


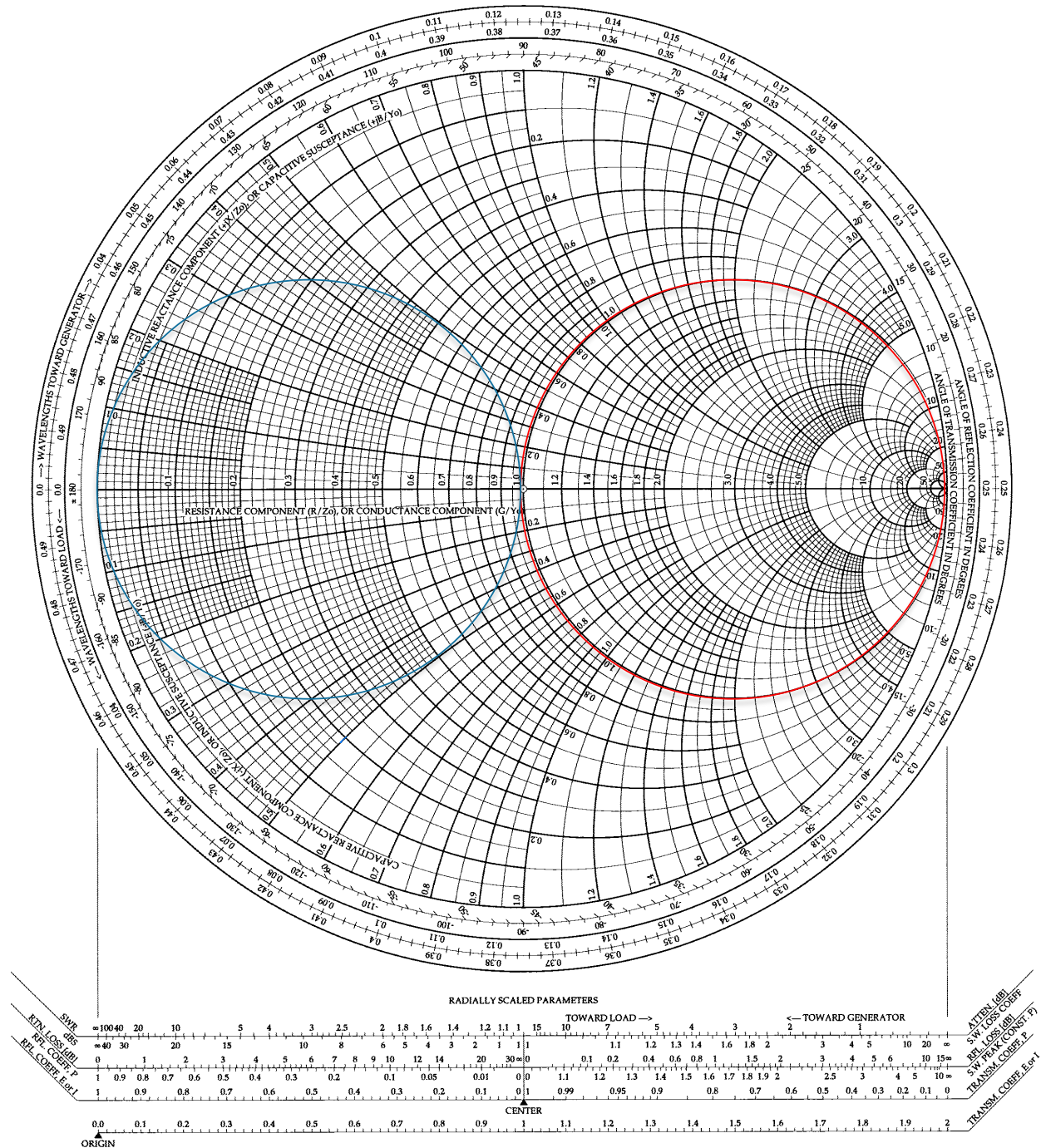
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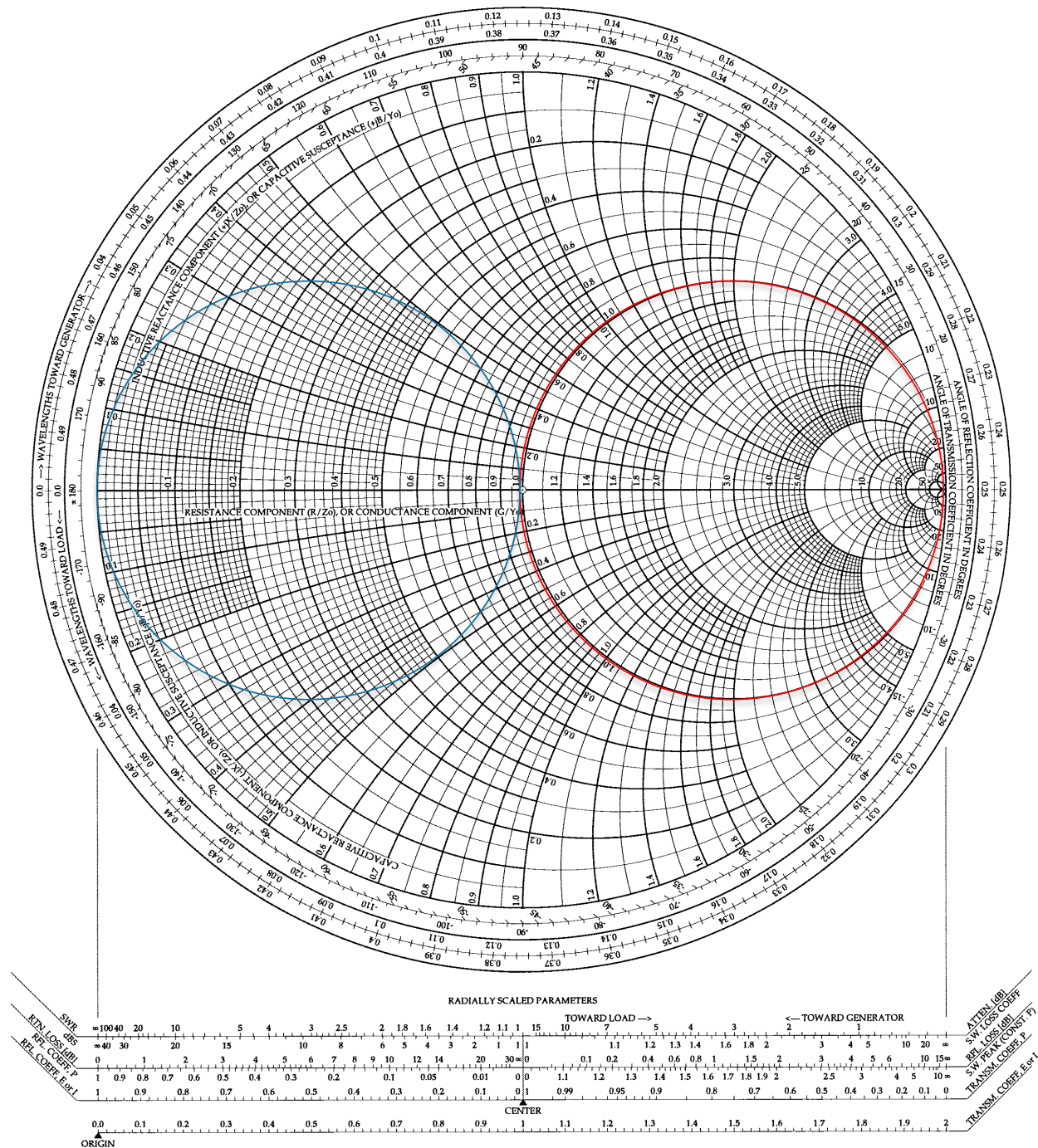
Matrl

Z_L

0.3 + 2.7j @ 200 MHz







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